ApexPlanet Cybersecurity Internship

Task 1: Foundation & Environment Setup

Internship Task: Task 1
Project: Cybersecurity Lab Setup and Basic Tool Practice

1. Objective

The objective of this task is to understand basic cybersecurity concepts, set up a virtual lab environment, practise Linux commands, test network connectivity, capture network packets, and perform basic cryptographic operations.

2. Lab Environment

The following tools and systems were used:

- **Operating System:** Kali Linux
- **Virtualization Software:** VMware / VirtualBox
- **Target Machine:** Metasploitable2
- **Network Analysis Tool:** Wireshark
- **Network Scanning Tool:** Nmap
- **Cryptography Tool:** OpenSSL

3. Virtual Lab Setup

Kali Linux and Metasploitable2 were prepared as virtual machines for cybersecurity practice. A private Host-only network was configured for lab communication.

The target machine was identified at IP address 192.168.18.129, and Kali Linux used IP address 192.168.18.128.

4. Network Connectivity Test

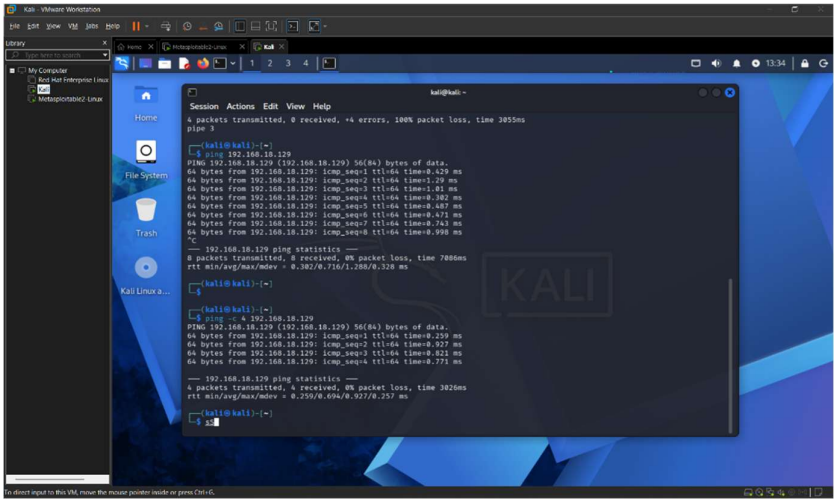
The ping command was used to test connectivity between Kali Linux and the target machine.

Command:

ping -c 4 192.168.18.129

The ping test completed successfully, confirming that the target responded to ICMP requests.

Evidence:

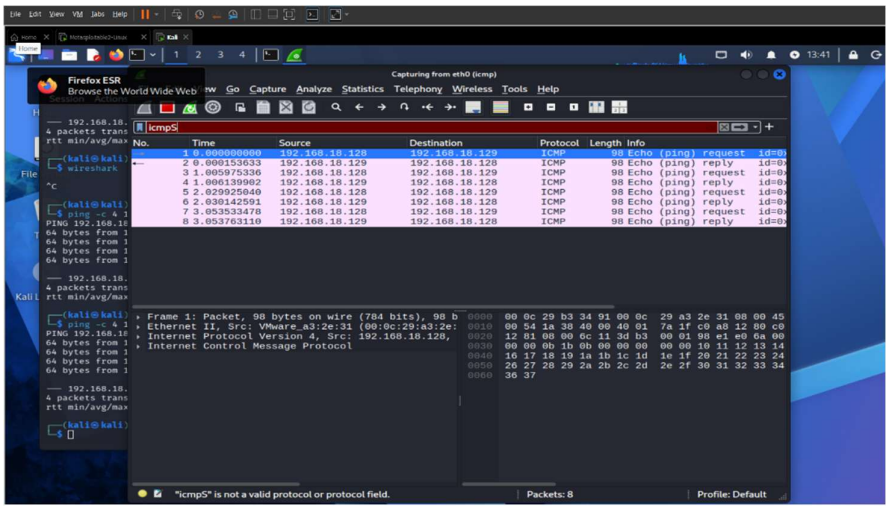


5. Wireshark Packet Capture

Wireshark was used to capture ICMP network traffic. The capture displayed Echo Request and Echo Reply packets between 192.168.18.128 and 192.168.18.129.

This exercise demonstrated how Wireshark can be used to observe and analyse network packets.

Evidence:

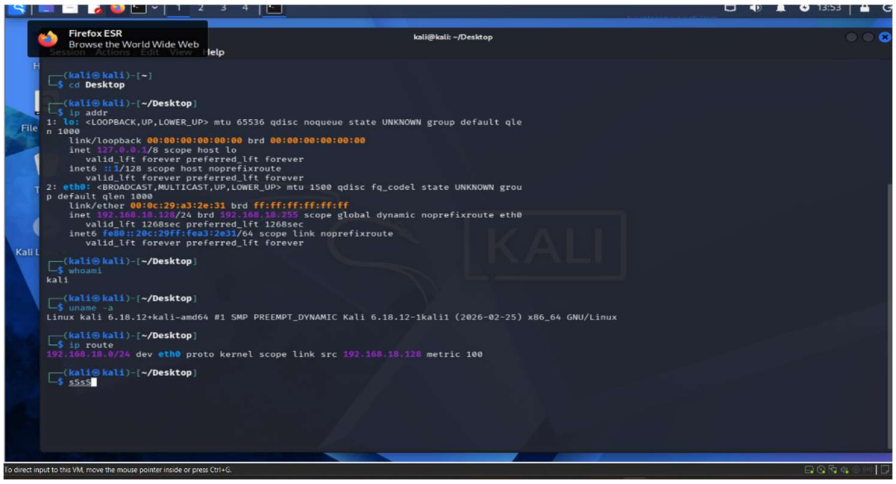


6. Linux Command Practice

The following Linux commands were practised:

- pwd — displays the current working directory.
- ll — lists files and directories in a detailed format.
- cd Desktop — changes the current directory to Desktop.
- ip addr — displays network interfaces and IP addresses.
- whoami — displays the current username.
- uname -a — displays Linux system information.
- ip route — displays routing information.

Evidence:



7. Cryptography Practice Using OpenSSL

OpenSSL was used to perform symmetric encryption and decryption using AES-256-CBC.

Encryption command:

openssl enc -aes-256-cbc -salt -pbkdf2 -in message.txt -out message.enc

Decryption command:

openssl enc -d -aes-256-cbc -pbkdf2 -in message.enc -out decrypted.txt

The decrypted output was verified using:

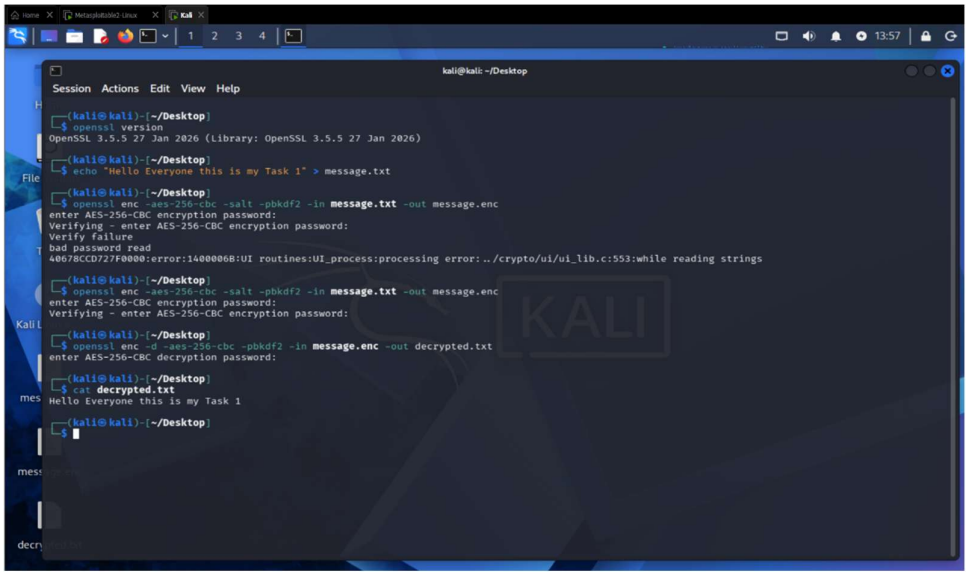
cat decrypted.txt

The original message was successfully recovered:

Hello Everyone this is my Task 1

This exercise demonstrated the basic process of encryption and decryption.

Evidence:



8. Nmap Network Scanning

Nmap was used to identify open TCP ports and detect services running on the Metasploitable2 lab machine.

Command:

nmap -sV 192.168.18.129

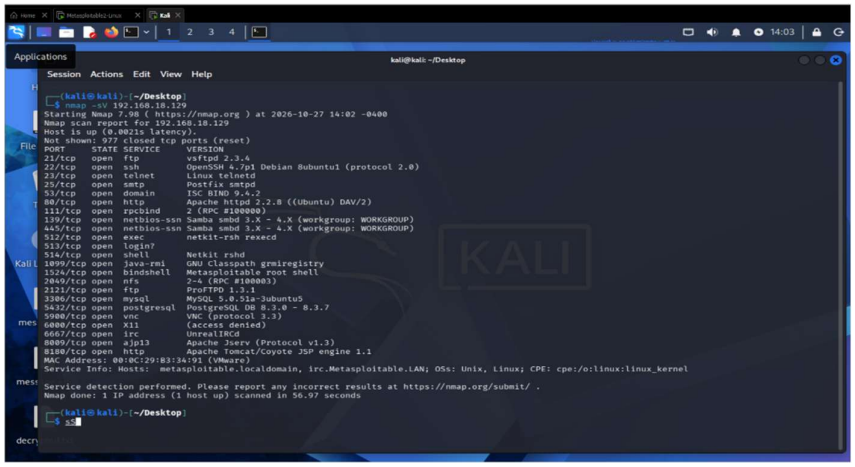
The scan reported that the target host was up and identified multiple open ports and services, including:

- Port 21 — FTP
- Port 22 — SSH

- Port 23 — Telnet
- Port 25 — SMTP
- Port 80 — HTTP
- Port 3306 — MySQL
- Port 5432 — PostgreSQL

The scan displayed 23 open TCP ports and detected several service versions.

Evidence:



9. Conclusion

This task provided practical experience with virtual lab setup, Linux commands, network connectivity testing, packet capture, basic cryptography, and network service scanning.

These activities helped build a foundation for further learning in cybersecurity and ethical hacking.